

Project Proposal (Proposed Solution)

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026

Team ID: SWTID-2026-3226

Project Title: FitBuddy – AI Fitness Plan Generator using Gemini Models

Maximum Marks: 5 Marks

Project Proposal (Proposed Solution) Report

FitBuddy is a Google Gemini powered fitness plan generator designed to help users turn their fitness goals into personalized workout plans, meal suggestions, weekly schedules, and progress guidance.

Project Overview

Objective	Provide an AI-assisted fitness workflow that turns a user's goals, fitness level, and preferences into a complete plan: generating workout routines, meal plans, weekly schedules, and motivational guidance.
Scope	Support common fitness-planning tasks through a web interface connected to a FastAPI backend and Google Gemini models.

Problem Statement

Description	Many people struggle to start or stay consistent with fitness because generic plans ignore their goals, fitness level, schedule, equipment, and dietary needs, while personal trainers and dietitians can be costly and information is scattered across many apps and websites.
Impact	These difficulties lead to lost motivation, unsafe or ineffective routines, and abandoned fitness goals. FitBuddy provides a single workflow from fitness goal to a personalized, structured plan.

Proposed Solution

Approach	Use Google Gemini models for personalized workout plan generation, meal and nutrition suggestions, weekly scheduling, exercise instructions, and motivational tips that adapt to each user's profile and goals.
Key Features	Goal and profile input; personalized workout plan; meal plan suggestions; weekly schedule; exercise descriptions and form tips; progress and motivation guidance.

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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Local development/testing on available CPU hardware; Gemini models run through the cloud API.
Memory	RAM	Sufficient memory for FastAPI and the Gemini client.
Storage	Application and project files	Project source, Python environment, configuration (.env), and generated fitness plans/results.

Software

Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries / APIs	AI and supporting libraries	Google GenAI SDK, python-dotenv, HTTPX
Development Environment	IDE / source control	VS Code or equivalent IDE; Git and GitHub

AI / Data

AI Services / Models	Model source and purpose	Google Gemini models for workout plan generation, meal suggestions, weekly scheduling, exercise guidance, and motivational tips.
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